Spatial Transcriptomics Analysis of Breast Cancer and Lymph Node Metastases

This repository presents a **reproducible, biologically grounded spatial transcriptomics analysis** of breast cancer progression, integrating 10x Visium data from **primary tumors** and **matched lymph node metastases (LNMT)**. The analysis combines **scRNA-seq–derived gene signatures** and **spatial scoring** to map epithelial, stromal, immune, and proliferative programs across tissues. By using **region-based annotation** instead of clustering, it provides a transparent, interpretable view of tumor–microenvironment architecture and how it reorganizes during metastasis. All steps—from preprocessing to interpretation—are contained in a single, fully commented Jupyter notebook, accompanied by a static PDF version.

Dataset Summary

Tissue Types: Primary breast tumors and matched lymph node metastases
Samples: 4 Visium slides (2 primary, 2 LNMT)
Signatures: Epithelial, Immune, Stromal, and Proliferation programs derived from prior scRNA-seq data

Modalities Integrated

- Spatial transcriptomics (10x Visium)
- Program scoring using curated gene sets
- Region-based annotation and masking
- Tissue classification (primary vs LNMT)
- Comparative analyses (composition, DE genes, spatial adjacency)

Analysis Overview

1. Sample Loading and Quality Control

- Four spatial datasets were loaded into `AnnData`.
- H&E overlays verified tissue coverage and morphology.
- Standard normalization and filtering steps ensured signal comparability.

2. Signature-Based Program Scoring

Each spot was scored for epithelial, stromal, immune, and proliferative gene programs using `sc.tl.score_genes()`.
Spatial plots revealed distinct localization patterns, highlighting intratumoral heterogeneity and compartmentalization.

3. Region-Based Annotation and Spatial Characterization

3.1 Region Masking via Signature Thresholds

Top-quantile masks (20%) defined high-activity regions for each program.
This approach prioritized interpretability and avoided overfitting to local clusters.

3.2 Combined Region Labeling

A hierarchical rule assigned one label per spot:
Tumor > Immune > Proliferative > Stroma > Unassigned.
The result is a mutually exclusive, biologically coherent tissue map.

3.3 Compact Region Plots

All slides were visualized in a consistent color scheme for easy cross-sample comparison of spatial architecture.

3.4 Region Composition Summary

Tables quantified the proportion of each region per tissue type:

- Primary tumors → higher epithelial and stromal content
- LNMT → greater immune and proliferative representation

3.5 Mean Signature Scores by Region

Averaged program scores confirmed biological specificity:
Tumor → epithelial enrichment   Immune → lymphoid activation   Stroma → ECM genes
Proliferation → cell cycle genes

3.6 Region–Region Spatial Adjacency

Adjacency graphs reconstructed neighborhood relations, revealing that **tumor and immune zones lie side-by-side but remain partially excluded**, a consistent spatial hallmark of breast cancer.

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4. Spatial Pattern Quantification

Region proportions and adjacency statistics were summarized across slides, demonstrating conserved **tumor–stroma boundaries** and **immune exclusion** phenomena.

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5. Tissue-Type Comparison: Primary vs LNMT

Comparative analyses showed that LNMT samples exhibit **immune and proliferative enrichment**, whereas primary tumors retain **epithelial and stromal dominance**.
These differences suggest metastatic sites are characterized by **immune activation and tissue remodeling**.

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6. Differential Expression Analysis

Differential expression between primary and LNMT samples yielded gene lists consistent with spatial findings.
Results highlight genes linked to **immune signaling, ECM remodeling, and proliferation**, serving as candidates for further validation.

Main Findings

- Epithelial, stromal, immune, and proliferative programs display **distinct, conserved spatial patterns**.
- **Tumor–immune compartmentalization** and stromal margins are reproducible features.
- Region-based annotation offers **interpretable functional maps** without clustering.
- LNMT tissues show **enhanced immune and proliferative activity**.
- Spatial adjacency captures **tissue-level coordination and exclusion mechanisms** in cancer.

Tools and Implementation

- **Languages / Libraries:** Python (Scanpy, Matplotlib, Pandas, NumPy)
- **Data Structure:** AnnData
- **Gene Scoring:** `sc.tl.score_genes()`
- **Environment:** Jupyter Notebook with inline figures and annotations
- **Output Organization:** `data/processed/`, `results/figures/`,

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results/metrics/
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Reproducibility and Notebook Details

The complete analysis is contained in:
`spatial_mapping.ipynb` — a fully commented Jupyter notebook where **code, results, and interpretations are integrated cell by cell**. A static, publication-quality version is available as **`docs/spatial_mapping.pdf`**, generated directly from the notebook.

Each step includes:

- Explanatory comments clarifying rationale and biological relevance
- Figures embedded inline
- Consistent foldering and output handling for reproducibility

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Limitations and Future Directions

While the current analysis provides interpretable and reproducible mapping of tumor–microenvironment structure, several extensions are planned:

- **Cross-dataset validation** to test whether region-based trends generalize to other cohorts (e.g., HER2+, TNBC).
- **Quantitative robustness analysis**, assessing how small changes in signature thresholds affect regional assignments.
- **Integration with graph-based frameworks** (e.g., SpatialMMKPNN) to model cell–cell communication and ligand–receptor signaling in spatial context.
- **Expanded visualization layer** with interactive overlays and 3D spatial projections.

These directions aim to transition this project from a single-study analysis to a reusable **benchmark for interpretable spatial transcriptomics workflows**.